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Investigation of Zn-diffusion in 2-micron InGaAs/GaAsSb superlattice planar diodes using atomic layer deposition of ZnO

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1. OUTLINE

Sensing and imaging applications such as light detection and ranging (LiDAR), and greenhouse gas sensing require a receiver with highly sensitive photodetectors such as avalanche photodiodes (APDs), which have high gain and speed.

Fabrication of APDs with planar architecture conventionally uses processes such as ion implantation, which can damage the material, and metal-organic chemical vapor deposition (MOCVD), which can be expensive and time-consuming.

We have explored Zn diffusion in epitaxially grown type-II InGaAs/GaAsSb SL on InP substrates. ZnO was grown with various thicknesses on the stack using ALD, and the Zn was diffused into the stack using a diffusion furnace under forming gas.

In this paper, we assess an alternative fabrication process to make APDs using Zn diffusion into epitaxially-grown material stacks through atomic layer deposition (ALD) of ZnO.

The effects of the diffusion process were investigated using various characterization techniques, including x-ray diffraction, and secondary ion mass spectroscopy.

The optimized Zn-diffusion process is then used to make planar diodes. Further optimization of this diffusion process may lead to a rapid, manufacturable, and cost-effective method of developing planar APDs

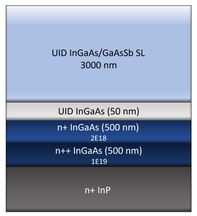
2. BACKGROUND

2.1 Approaches for dopant incorporation in planar diodes

MOCVD	Ion Implantation	Thermal Diffusion	Proposed Method: ALD deposited ZnO
Pros: • Diffusion & doping is precise and accurate • Maintains material quality **Cons**: • Expensive • Time-consuming	**Pros**: • Diffusion & doping is precise and accurate **Cons**: • Damages the material • Expensive • Time-consuming	**Pros**: • Maintains material quality **Cons**: • Complex process	**Pros**: • Maintains material quality • Simple process • Fast **Cons** • Not researched enough

2.2 Material of Choice – InGaAs/GaAsSb SL

InGaAs/GaAsSb SL: Optical response up to 2.5 µm*
- Competitor to HgCdTe in the SWIR region due to its low dark current*



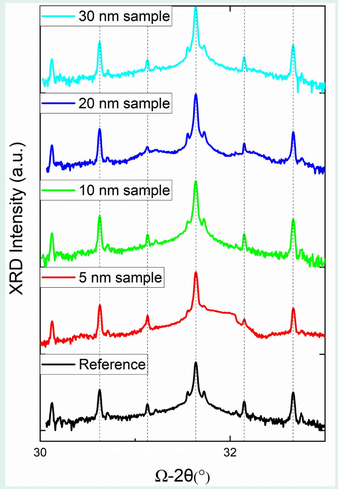
As grown material

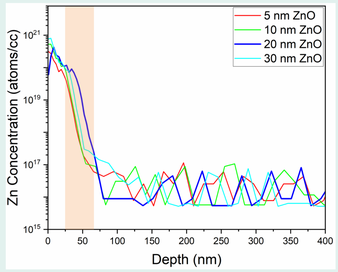
Material	Doping	Thickness
InGaAs/GaAsSb SL	UID	3000 nm
InGaAs	UID	50 nm
InGaAs	n^+: 2 x 10^{18}	500 nm
InGaAs	n^{++}: 1 x 10^{19}	500nm
InP Substrate		

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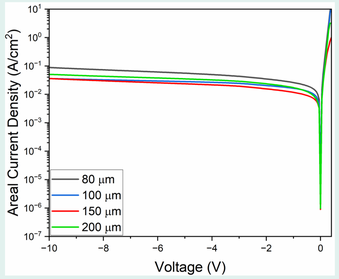
KEY RESULTS

First Zn diffusion in InGaAs/GaAsSb SL using ALD deposited ZnO

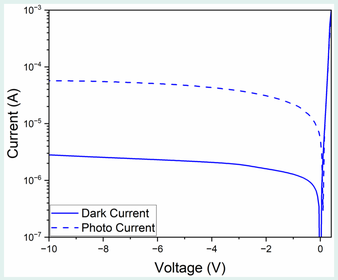




- XRD measurement states **material quality is maintained** after the Zn diffusion is complete
- SIMS measurement confirmed the **diffusion depth of ~50 nm**, irrespective of the deposited ZnO thickness



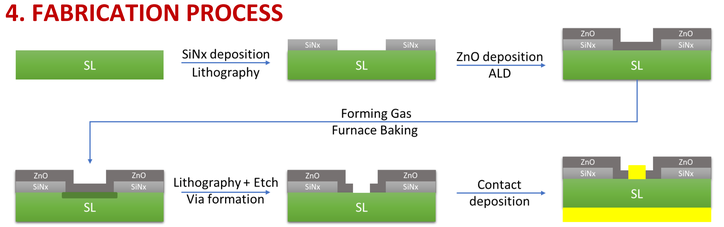
- Dark **current measurement confirmed the diffusion of Zn** from ZnO layer to form the p-region
- Current density analysis confirmed current leakage



- Photocurrent measurements were performed on a device with 100 µm diameter, with a 2-µm wavelength laser to **confirm the operation of the diode at 2-µm**

3. DIFFUSION PROCESS

4. FABRICATION PROCESS

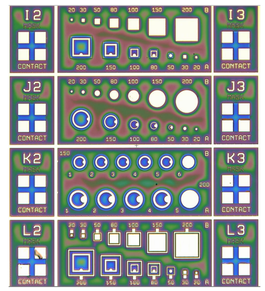


5. CONCLUSION AND FUTURE WORK

- **Achieved Zn diffusion** through the proposed method
- **Characterized the material** and devices to show material quality is maintained but the dark current is high due to current leakage

Future Work:
- To design and demonstrate Zn-diffused APD with separate absorption charge and multiplication layer (SACM structure)
- To test devices with a guard ring structure

Fabricated devices



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Reference Paper

SPIE paper number 12534-8: Investigation of Zn-diffusion in 2-micron InGaAs/GaAsSb superlattice planar diodes using atomic layer deposition of ZnO